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Statistics

STAT101 | Suggested sequence · Semester 1 | Student-authored study guide

How to use these notes

Descriptive statistics, sampling, estimation, confidence intervals, tests, and responsible interpretation. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Describing distributions

Summarize center with mean or median and spread with standard deviation or IQR. Inspect shape and outliers visually. A statistic describes a sample; it does not automatically establish a population fact.

Worked example

```
For incomes with a long right tail, report median and IQR alongside a histogram.
```

Check your understanding

- When is the median robust?
- What does standard deviation measure?

MODULE 02 · Sampling and uncertainty

A sample should represent the target population for generalization. Random sampling reduces selection bias; larger samples generally reduce standard error but do not repair biased collection. Confidence intervals describe a procedure’s long-run coverage.

Worked example

```
A 95% confidence procedure covers the fixed population parameter in about 95% of repeated
samples under its assumptions.
```

Check your understanding

- Does a larger biased sample remove bias?
- How should a 95% interval be interpreted?

MODULE 03 · Tests, effect sizes, and association

A p-value measures how surprising data are under a null model; it is not the probability the hypothesis is true. Report effect size and uncertainty. Correlation describes association, not causal effect.

Worked example

```
Report a mean difference with a confidence interval instead of only “p < .05”.
```

Check your understanding

- What is a Type I error?
- Why does correlation not establish causation?

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.